

MACHINE LEARNING INTERNSHIP TASK 1

Housing Price Predictor Model

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Data Exploration & Insight Analysis

DATASET CHARACTERISTICS

The **California Housing Dataset** provides a robust baseline for regression analysis, focusing on neighborhood-level statistics.

- **Observation Count:** 20,640 unique district records.
- **Feature Integrity:** Zero missing values (Null-free).
- **Primary Drivers:** Median Income and geographical coordinates are strongly correlated with value.



PRIMARY PERFORMANCE METRIC

0.576

R-Squared Score

Our Linear Regression baseline successfully captures **57.6%** of the variance in housing prices. This represents a solid foundational performance for an unoptimized model.

MODEL EVALUATION METRICS

<i>Metric</i>	<i>Value</i>	<i>Interpretation</i>
<i>MAE</i>	<i>0.533</i>	<i>Average prediction error of ~\$53,300</i>
<i>RMSE</i>	<i>0.746</i>	<i>Impact of pricing outliers in the dataset</i>
<i>R² Score</i>	<i>0.576</i>	<i>Variance explained by neighborhood features</i>

FUTURE STRATEGIC ROADMAP



Feature Engineering

Developing interaction terms like 'Rooms per Household' to refine feature density.



Algorithm Shift

Transitioning to Random Forest or Gradient Boosting to handle non-linear dependencies.



Geospatial Clusters

Implementing K-Means clustering on coordinates to capture local market premiums.

Thank You

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